

Re-ERP Inventory Management Module

Requirements & Design Document (RD) - v2.7 - September 2026

Prepared for the Re-ERP product team. Defines the data model, web services and screens for the Inventory Management module, and the roadmap toward Purchasing, Order Management and Cost Accounting.

1. Executive Summary

This document is the complete functional and technical design for the Re-ERP Inventory Management module. Version 2.0 extends the v1.0 draft to the full footprint of a modern enterprise inventory system: the organization hierarchy and every setup object, a deep item master with grouped attributes, categories and catalogs, revisions and attachments, lot and serial control, the transaction engine with reasons and account aliases, reservations and availability, inter-organization transfers and transfer orders, cycle counting, physical inventory and ABC analysis, cost organizations with cost periods and period close, on-hand and valuation inquiry, plus the complete web-service catalog and screen designs. Purchasing, Order Management and full Cost Accounting remain separate modules built on the hooks reserved here.

1.1 Functional coverage map

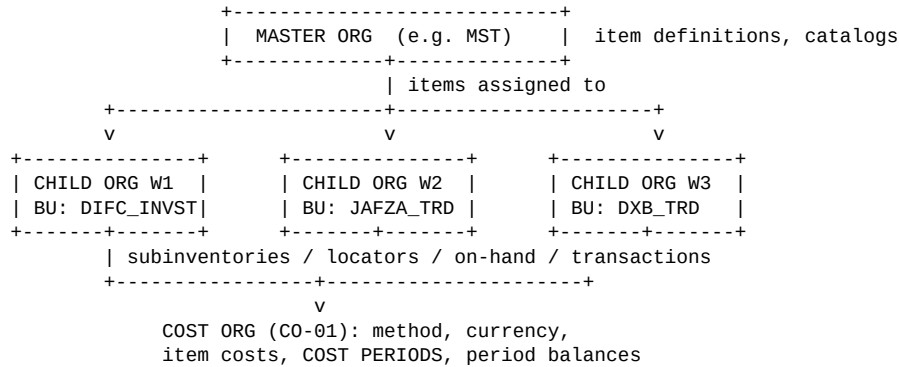
Area	Contents	Section
Organizations	Master/child orgs, BU assignment, org access control	2, 4
Setup objects	Parameters (inventory/receiving/shipping), subinventories, locators, UOMs, reasons, account aliases, material statuses, shipping methods	4, 8
Items	Attribute groups (main, inventory, costing, purchasing, sales, physical, planning), statuses, categories & catalogs, revisions, attachments, org assignments with overrides	5
Lot & serial	Lot master, serial master, transaction detail capture	6
Costing	Cost organizations, item costs, cost periods, period close, period balances	7
Transactions	Engine, types, validations, accounting hand-off	8
Reservations	Demand-linked reservations, available-to-transact	9
Transfers	Direct & in-transit inter-org, transfer orders with lifecycle	10
Counting	Cycle counting, physical inventory, ABC analysis	11
Inquiry	On-hand, availability, valuation, transaction history	12
Services & UI	ORDS catalog with payloads; screen designs	13, 14
Rollout	Numbering, seeding, phases, open decisions	15-17

1.2 Out of scope (separate modules)

- Purchasing: requisitions, purchase orders, receiving against orders - Phase 2 (receipt routing, tolerances and the PO_RECEIPT transaction type are reserved here)
- Order Management: sales orders, picking waves, ship confirm - Phase 3 (SO_ISSUE reserved; reservations in section 9 already support sales-order demand)
- Cost Accounting: perpetual average recalculation, standard cost revisions, variance analysis - Phase 4 (cost periods and cost engine, elements, adjustments and periods in section 7 are its foundation)
- Manufacturing/WIP, warehouse task management, advanced picking rules

2. Organization Model

One organization per company is the ITEM MASTER organization where items are defined. Child organizations are the physical warehouses/plants that hold stock and transact; each child is linked to its master, assigned to a Business Unit (existing gl/businessunits data) and inherits that BU's legal entity. An item becomes usable in a child organization only through an item-organization assignment. Cost organizations group inventory organizations for valuation: costing method, costing currency, item costs and cost periods all live at cost-organization level.



2.1 Organization access control

RR_INV_ORG_ACCESS restricts which application users can transact in which organizations. Every service that writes stock movements checks it; inquiry pages filter their org LOV by it. A user with no rows has no inventory access; an ALL_ORGS flag supports administrators.

Column	Type	Req	Description
ACCESS_ID	NUMBER	Y	PK
USER_NAME	VARCHAR2(150)	Y	Re-ERP login (erp_user)
ORGANIZATION_ID	NUMBER	N	NULL together with ALL_ORGS_FLAG = Y
ALL_ORGS_FLAG	VARCHAR2(1)	Y	Y grants every organization (default N)
Audit columns	-	Y	-

3. Entity Map

Parent	Child	Link	Cardinality
RR_INV_ORGANIZATIONS (master)	RR_INV_ORGANIZATIONS (child)	MASTER_ORG_ID	1 : N
GL business units (existing)	RR_INV_ORGANIZATIONS	BUSINESS_UNIT_NAME	1 : N
RR_INV_ORGANIZATIONS	RR_INV_ORG_PARAMETERS / RECEIVING / SHIPPING	ORGANIZATION_ID	1 : 1 each
RR_INV_ORGANIZATIONS	RR_INV_SUBINVENTORIES	ORGANIZATION_ID	1 : N
RR_INV_SUBINVENTORIES	RR_INV_LOCATORS	SUBINVENTORY_ID	1 : N
RR_INV_ORGANIZATIONS (master)	RR_INV_ITEMS	MASTER_ORG_ID	1 : N
RR_INV_ITEMS	RR_INV_ITEM_ORG_ASSIGNMENTS	ITEM_ID	1 : N
RR_INV_ITEMS	RR_INV_ITEM_REVISIONS / RR_INV_ITEM_ATTACHMENTS	ITEM_ID	1 : N
RR_INV_CATEGORY_SETS	RR_INV_CATEGORIES	CATEGORY_SET_ID	1 : N
RR_INV_CATEGORIES	RR_INV_ITEM_CATEGORIES	CATEGORY_ID	1 : N (one per item per set)
RR_INV_ITEMS	RR_INV_LOTS / RR_INV_SERIALS	ITEM_ID	1 : N
RR_INV_COST_ORGANIZATIONS	RR_INV_COST_ORG_ASSIGNMENTS	COST_ORG_ID	1 : N (inv org unique)
RR_INV_COST_ORGANIZATIONS	RR_INV_ITEM_COSTS / RR_INV_COST_PERIODS	COST_ORG_ID	1 : N
RR_INV_COST_PERIODS	RR_INV_PERIOD_BALANCES	COST_PERIOD_ID	1 : N
RR_INV_TXN_TYPES / REASONS	RR_INV_TRANSACTIONS	codes	1 : N

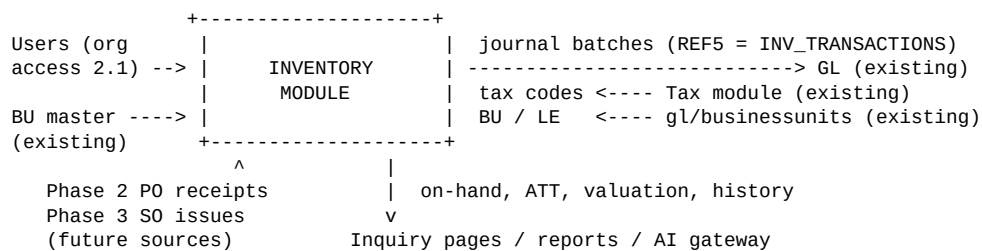
Parent	Child	Link	Cardinality
RR_INV_TRANSACTIONS	RR_INV_TXN_LOT_DETAILS / TXN_SERIAL_DETAILS	TRANSACTION_ID	1 : N
RR_INV_TRANSACTIONS	RR_INV_TXN_ACCOUNTING	TRANSACTION_ID	1 : N (Dr/Cr)
org+subinv+locator+item(+lot)	RR_INV_ONHAND_BALANCES	composite unique	1 row per combination
RR_INV_ORGANIZATIONS	RR_INV_RESERVATIONS	ORGANIZATION_ID	1 : N
RR_INV_TRANSFER_ORDERS	RR_INV_TRANSFER_ORDER_LINES	TRANSFER_ORDER_ID	1 : N
RR_INV_INTERORG_SHIPMENTS	RR_INV_TRANSACTIONS	SHIPMENT_ID	1 : N
RR_INV_CYCLE_COUNTS	RR_INV_CYCLE_COUNT_ENTRIES	CYCLE_COUNT_ID	1 : N
RR_INV_PHYSICAL_INVENTORIES	RR_INV_PHYS_INV_TAGS	PHYSICAL_INV_ID	1 : N
RR_INV_ABC_COMPILES	RR_INV_ABC_ASSIGNMENTS	COMPILE_ID	1 : N

All primary keys come from one sequence RR_INV_SEQ starting at 5000000001.

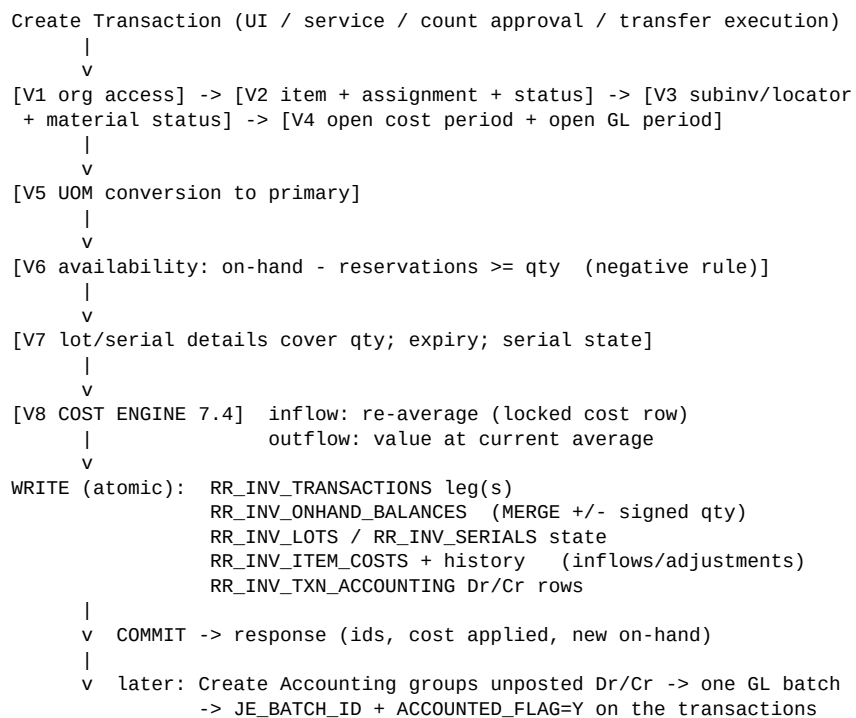
WHO columns: every RR_INV_* table carries the standard audit (WHO) columns - CREATED_BY VARCHAR2(150), CREATION_DATE TIMESTAMP DEFAULT SYSTIMESTAMP, LAST_UPDATED_BY VARCHAR2(150), LAST_UPDATE_DATE TIMESTAMP, LAST_UPDATE_LOGIN VARCHAR2(100). The packages stamp them from the p_user parameter on every insert and update; immutable ledgers (transactions, cost history, execution log, period balances, snapshots) fill them at insert only and are never updated. Where a table section below says 'Audit columns', it means exactly this set.

3.2 Data flow diagrams

Level 0 - context



Level 1 - transaction processing (one database transaction)



Level 1 - inter-org transfer

Transfer Order (optional document) Ad-hoc transfer

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DRAFT->REQUESTED->APPROVED |
\----- ship -----+
|
SOURCE ORG: INTERORG_SHIP / DIRECT_OUT leg (-qty at source avg)
| shipment row (SHIPPED, carries cost)
v
[in-transit bucket visible in inquiry]
|
DEST ORG: receive -> INTERORG_RECEIVE / DIRECT_IN leg (+qty)
| cross cost org? -> re-average at carried cost (7.6)
v
shipment RECEIVED -> TO lines roll up -> COMPLETED

```

Level 1 - counting and period close

```

Cycle count def / Physical inventory (user-created)
| generate entries / snapshot (PI freezes scope)
v
count entry -> variance vs snapshot -> tolerance / approval
| |
v v
no variance: close entry CYCLE_COUNT_ADJ / PHYS_INV_ADJ
                        through the SAME engine pipeline above
Period close (per cost org):
[no unaccounted txns] + [no open PI] + [in-transit warning]
-> snapshot on-hand x avg into RR_INV_PERIOD_BALANCES -> CLOSED

```

3.3 Complete database object inventory

Tables (all with WHO columns)

Table	Purpose
RR_INV_ORGANIZATIONS	Master/child inventory organizations
RR_INV_ORG_PARAMETERS	Inventory parameters (1:1 org)
RR_INV_RECEIVING_PARAMETERS	Receiving parameters (1:1 org)
RR_INV_SHIPPING_PARAMETERS	Shipping parameters (1:1 org)
RR_INV_ORG_ACCESS	User to organization security
RR_INV_SUBINVENTORIES	Subinventories per org
RR_INV_LOCATORS	Locators per subinventory
RR_INV_UOMS	Units of measure
RR_INV_UOM_ITEM_CONVERSIONS	Item-specific UOM conversions
RR_INV_TXN_TYPES	Seeded + user-defined transaction types
RR_INV_TXN_REASONS	Transaction reasons
RR_INV_ACCOUNT_ALIASES	Named account destinations per org
RR_INV_MATERIAL_STATUSES	Location/lot usage restrictions
RR_INV_SHIP_METHODS	Carriers / ship methods
RR_INV_ITEM_STATUSES	Item status templates
RR_INV_ITEMS	Item master (all attribute groups)
RR_INV_ITEM_ORG_ASSIGNMENTS	Item availability + overrides per org
RR_INV_ITEM_REVISIONS	Item revisions
RR_INV_ITEM_ATTACHMENTS	Item files (BLOB, standard mechanism)
RR_INV_CATEGORY_SETS / RR_INV_CATEGORIES / RR_INV_ITEM_CATEGORIES	Catalogs and item categorization
RR_INV_LOTS	Lot master per item + org
RR_INV_SERIALS	Serial master per item
RR_INV_COST_ORGANIZATIONS	Cost organizations
RR_INV_COST_ORG_ASSIGNMENTS	Inventory org to cost org (unique)
RR_INV_COST_ELEMENTS	Cost element lookup
RR_INV_ITEM_COSTS	Current average per cost org + item
RR_INV_ITEM_COST_DETAILS	Element breakdown of the current cost
RR_INV_ITEM_COST_HISTORY	Every average movement (immutable)
RR_INV_COST_ADJUSTMENTS	Cost adjustment register
RR_INV_COST_PERIODS	Cost periods per cost org
RR_INV_PERIOD_BALANCES	Period-end valuation snapshots (immutable)
RR_INV_TRANSACTIONS	Material transaction ledger (immutable)
RR_INV_TXN_LOT_DETAILS / RR_INV_TXN_SERIAL_DETAILS	Lot/serial capture per leg
RR_INV_TXN_ACCOUNTING	Dr/Cr distribution rows per transaction
RR_INV_ONHAND_BALANCES	Maintained running totals (MERGE +/-)
RR_INV_RESERVATIONS	Demand-linked reservations
RR_INV_INTERORG_SHIPMENTS	Transfer execution documents
RR_INV_TRANSFER_ORDERS / RR_INV_TRANSFER_ORDER_LINES	Requested transfers with lifecycle
RR_INV_CYCLE_COUNTS / RR_INV_CYCLE_COUNT_ITEMS / RR_INV_CYCLE_COUNT_ENTRIES	Cycle counting
RR_INV_PHYSICAL_INVENTORIES / RR_INV_PHYS_INV_SNAPSHOT / RR_INV_PHYS_INV_TAGS	Physical inventory
RR_INV_ABC_COMPILES / RR_INV_ABC_CLASSES / RR_INV_ABC_ASSIGNMENTS	ABC analysis

Table	Purpose
RR_INV_PROC_REGISTRY	Whitelist of callable procedures (13.2)
RR_INV_EXEC_LOG	Every gateway call: user, proc, params, status, ms (immutable)

Views (read path, section 13.1)

View	Purpose
RR_INV_V_ORGS	Org grid with BU/LE and parameter highlights
RR_INV_V_SUBINVENTORIES	Subinventories + locator counts + on-hand value
RR_INV_V_ITEMS	Item grid: flattened attribute groups, category, assigned orgs
RR_INV_V_ITEM_COSTS	Current avg, basis, element breakdown
RR_INV_V_COST_HISTORY	Average movements with transaction refs
RR_INV_V_ONHAND	On-hand / reserved / ATT / in-transit by level
RR_INV_V_TRANSACTIONS	History incl. accounting join and GL batch
RR_INV_V_RESERVATIONS	Active + history with fulfilment
RR_INV_V_SHIPMENTS	Shipments with lines and status
RR_INV_V_TRANSFER_ORDERS	TO headers + line fulfilment roll-up
RR_INV_V_COUNT_ENTRIES	Cycle count worklist with variance calc
RR_INV_V_PHYS_TAGS	Physical inventory tags vs snapshot
RR_INV_V_COST_PERIODS	Periods with close-blocker counts
RR_INV_V_PERIOD_BALANCES	Valuation snapshots
RR_INV_V_EXPIRING_LOTS	Lots by days-to-expiry

Other objects

- Sequence RR_INV_SEQ (all PKs)
- Packages: the eight RR_INV_*_PKG packages specified in 13.3
- ORDS: ONE dispatcher endpoint inv/execute + the item-attachment endpoints; reads use the existing ai/executequery
- Seed script: UOMs, txn types, item statuses, material statuses, reasons, cost elements, category sets, ship methods

4. Core Setup Objects

4.1 RR_INV_ORGANIZATIONS

Column	Type	Req	Description
ORGANIZATION_ID	NUMBER	Y	PK (RR_INV_SEQ)
ORGANIZATION_CODE	VARCHAR2(18)	Y	Short unique code (MST, W1, W2)
ORGANIZATION_NAME	VARCHAR2(240)	Y	Display name
ORG_TYPE	VARCHAR2(10)	Y	MASTER or CHILD
MASTER_ORG_ID	NUMBER	child	FK to master; NULL on master rows
BUSINESS_UNIT_NAME	VARCHAR2(240)	child	Assigned BU
LEGAL_ENTITY_NAME	VARCHAR2(360)	N	Copied from BU at assignment
LOCATION	VARCHAR2(240)	N	Site / address
STATUS	VARCHAR2(10)	Y	ACTIVE / INACTIVE
Audit columns	-	Y	CREATED_BY, CREATION_DATE, LAST_UPDATED_BY, LAST_UPDATE_DATE

Rules: UNIQUE(ORGANIZATION_CODE); a CHILD must carry MASTER_ORG_ID and BUSINESS_UNIT_NAME; an organization with on-hand or open documents cannot be deactivated; the master org is setup-only (no transactions) unless the open decision in section 17 says otherwise.

4.2 RR_INV_ORG_PARAMETERS (1:1)

Column	Type	Req	Description
ORGANIZATION_ID	NUMBER	Y	PK + FK
PRIMARY_UOM_CODE	VARCHAR2(10)	Y	Default UOM (EA)
DEFAULT_SUBINVENTORY	VARCHAR2(30)	N	Default receiving subinventory
LOCATOR_CONTROL	VARCHAR2(15)	Y	NONE / PRESPECIFIED
NEGATIVE_INV_ALLOWED_FLAG	VARCHAR2(1)	Y	Allow negative on-hand (default N)
LOT_CONTROL_ENABLED_FLAG	VARCHAR2(1)	Y	Org-level enablement for lot capture
SERIAL_CONTROL_ENABLED_FLAG	VARCHAR2(1)	Y	Org-level enablement for serial capture
AUTO_CREATE_LOTS_FLAG	VARCHAR2(1)	Y	Create lot master rows on first receipt (default Y)
CURRENCY_CODE	VARCHAR2(15)	Y	Valuation currency (default AED)
MATERIAL_ACCOUNT	VARCHAR2(500)	Y	Inventory valuation account
ADJUSTMENT_ACCOUNT	VARCHAR2(500)	Y	Misc receipt/issue offset
CYCLE_COUNT_ADJ_ACCOUNT	VARCHAR2(500)	N	Count adjustment offset (falls back to adjustment a/c)
PHYS_INV_ADJ_ACCOUNT	VARCHAR2(500)	N	Physical inventory adjustment offset
INTERORG_RECEIVABLE_ACCOUNT	VARCHAR2(500)	N	Inter-org receivable
INTERORG_PAYABLE_ACCOUNT	VARCHAR2(500)	N	Inter-org payable
INTRANSIT_ACCOUNT	VARCHAR2(500)	N	In-transit inventory
Audit columns	-	Y	-

Accounts are full code combinations picked with the existing AccountSelector, company segment locked from the BU (the Manage Receipts pattern).

4.3 RR_INV_RECEIVING_PARAMETERS (1:1)

Column	Type	Req	Description
ORGANIZATION_ID	NUMBER	Y	PK + FK
RECEIPT_ROUTING	VARCHAR2(15)	Y	DIRECT / STANDARD / INSPECTION (default DIRECT)
RECEIPT_NUM_GENERATION	VARCHAR2(10)	Y	AUTOMATIC / MANUAL
NEXT_RECEIPT_NUMBER	NUMBER	Y	Counter for AUTOMATIC
OVER_RECEIPT_TOLERANCE_PERCENT	NUMBER	N	Used by Phase 2 PO receiving
OVER_RECEIPT_ACTION	VARCHAR2(10)	N	NONE / WARNING / REJECT
EARLY_RECEIPT_DAYS / LATE_RECEIPT_DAYS	NUMBER	N	Date tolerances (Phase 2)
ALLOW_SUBSTITUTE_RECEIPTS_FLAG	VARCHAR2(1)	N	Phase 2
ALLOW_UNORDERED_RECEIPTS_FLAG	VARCHAR2(1)	N	Phase 2
RECEIVING_INVENTORY_ACCOUNT	VARCHAR2(500)	N	Receiving clearing
Audit columns	-	Y	-

4.4 RR_INV_SHIPPING_PARAMETERS (1:1)

Column	Type	Req	Description
ORGANIZATION_ID	NUMBER	Y	PK + FK
SHIPMENT_NUM_GENERATION	VARCHAR2(10)	Y	AUTOMATIC / MANUAL
NEXT_SHIPMENT_NUMBER	NUMBER	Y	Counter
DEFAULT_SHIP_METHOD	VARCHAR2(60)	N	FK to shipping methods lookup
PICK_CONFIRM_REQUIRED_FLAG	VARCHAR2(1)	N	Phase 3
AUTO_RECEIVE_DIRECT_FLAG	VARCHAR2(1)	Y	Direct transfers auto-complete (default Y)

Column	Type	Req	Description
GOODS_IN_TRANSIT_ACCOUNT	VARCHAR2(500)	N	In-transit shipping account
Audit columns	-	Y	-

4.5 RR_INV_SUBINVENTORIES

Column	Type	Req	Description
SUBINVENTORY_ID	NUMBER	Y	PK
ORGANIZATION_ID	NUMBER	Y	FK
SUBINVENTORY_CODE	VARCHAR2(30)	Y	Unique within org (STORES, FG, DAMAGED, QA)
DESCRIPTION	VARCHAR2(240)	N	-
SUBINV_TYPE	VARCHAR2(15)	Y	STORAGE / RECEIVING
LOCATOR_CONTROL	VARCHAR2(15)	Y	NONE / PRESPECIFIED (defaults from org)
MATERIAL_STATUS_CODE	VARCHAR2(30)	N	FK to material statuses (4.9); NULL = unrestricted
MATERIAL_ACCOUNT	VARCHAR2(500)	N	Overrides org material account
ASSET_FLAG	VARCHAR2(1)	Y	Y valued stock / N expense subinventory
QUANTITY_TRACKED_FLAG	VARCHAR2(1)	Y	N = no on-hand kept (expense areas)
PICKING_ORDER	NUMBER	N	Sort order for issue suggestions
STATUS	VARCHAR2(10)	Y	ACTIVE / INACTIVE
Audit columns	-	Y	-

4.6 RR_INV_LOCATORS

Column	Type	Req	Description
LOCATOR_ID	NUMBER	Y	PK
SUBINVENTORY_ID	NUMBER	Y	FK
LOCATOR_CODE	VARCHAR2(60)	Y	Row-Rack-Bin style (A1-R2-B3), unique in subinventory
DESCRIPTION	VARCHAR2(240)	N	-
MATERIAL_STATUS_CODE	VARCHAR2(30)	N	Optional restriction (4.9)
PICKING_ORDER	NUMBER	N	Sort order for issue suggestions
CAPACITY_QTY / CAPACITY_UOM	NUMBER / VARCHAR2(10)	N	Informational capacity
STATUS	VARCHAR2(10)	Y	ACTIVE / INACTIVE

4.7 RR_INV_UOMS and conversions

Column	Type	Req	Description
UOM_CODE	VARCHAR2(10)	Y	PK (EA, KG, G, M, CM, L, ML, BOX, PKT, ROLL, DOZ, SET)
UOM_NAME	VARCHAR2(60)	Y	Each, Kilogram...
UOM_CLASS	VARCHAR2(30)	Y	QUANTITY / WEIGHT / LENGTH / VOLUME
BASE_FLAG	VARCHAR2(1)	Y	Y = class base
CONVERSION_TO_BASE	NUMBER	N	Standard conversion to class base

RR_INV_UOM_ITEM_CONVERSIONS (ITEM_ID, UOM_CODE, CONVERSION_TO_PRIMARY) allows item-specific conversions (e.g. 1 BOX = 24 EA for one item, 12 EA for another). Transactions in Phase 1 capture the entered UOM and store quantity converted to primary UOM.

4.8 RR_INV_TXN_REASONS and RR_INV_ACCOUNT_ALIASES

Transaction reasons are a controlled list (DAMAGE, EXPIRY, SAMPLE, OPENING, CORRECTION, DONATION ...) selectable on any transaction; columns: REASON_CODE PK, REASON_NAME, DESCRIPTION, STATUS. Account aliases give a named account destination for miscellaneous transactions (alias MARKETING-SAMPLES points at a specific expense combination): ALIAS_ID PK, ORGANIZATION_ID, ALIAS_CODE unique per org, ACCOUNT_COMBINATION, STATUS. A miscellaneous receipt/issue may specify an alias; its account then replaces the adjustment account on the offset line.

4.9 RR_INV_MATERIAL_STATUSES

Named statuses that restrict what can happen to stock in a subinventory, locator or lot. Columns: STATUS_CODE PK, STATUS_NAME, ALLOW_RECEIPT_FLAG, ALLOW_ISSUE_FLAG, ALLOW_TRANSFER_FLAG, ALLOW_RESERVATION_FLAG, DESCRIPTION. Seeded: UNRESTRICTED, HOLD (no issues/transfers), QUARANTINE (receipts only), REJECTED (transfers only). The transaction engine checks the effective status of source and destination.

4.10 RR_INV_SHIP_METHODS

Lookup of carriers/methods used by shipping parameters and transfer orders: SHIP_METHOD_CODE PK, NAME, STATUS.

5. Items - Full Item Master

Items are created in the master organization and carry their attributes in functional groups. Attribute values marked 'org-overridable' can be overridden per child organization on the assignment row; everything else is master-level only.

5.1 RR_INV_ITEMS - attribute groups

Group: Main

Column	Type	Req	Description
ITEM_ID	NUMBER	Y	PK
MASTER_ORG_ID	NUMBER	Y	FK to master org
ITEM_NUMBER	VARCHAR2(80)	Y	Unique item code
DESCRIPTION	VARCHAR2(240)	Y	Short description
LONG_DESCRIPTION	VARCHAR2(2000)	N	-
ITEM_TYPE	VARCHAR2(30)	Y	FINISHED_GOOD / PURCHASED / RAW_MATERIAL / CONSUMABLE / SERVICE / OTHER
ITEM_STATUS_CODE	VARCHAR2(30)	Y	FK to item statuses (5.2), default ACTIVE
PRIMARY_UOM_CODE	VARCHAR2(10)	Y	FK to UOMs
CURRENT_REVISION	VARCHAR2(10)	N	Latest revision label (5.5)

Group: Inventory

Column	Type	Req	Description
STOCKABLE_FLAG	VARCHAR2(1)	Y	Can be stocked (default Y)
TRANSACTABLE_FLAG	VARCHAR2(1)	Y	Can transact (default Y)
RESERVABLE_FLAG	VARCHAR2(1)	Y	Can be reserved (default Y)
LOT_CONTROL_FLAG	VARCHAR2(1)	Y	Lot number required on transactions (default N)
LOT_EXPIRATION_CONTROL_FLAG	VARCHAR2(1)	Y	Lots carry expiry dates (default N)
SHELF_LIFE_DAYS	NUMBER	N	Default expiry = receipt date + days
SERIAL_CONTROL_FLAG	VARCHAR2(1)	Y	Serial per unit required (default N)
LOT_PREFIX / SERIAL_PREFIX	VARCHAR2(20)	N	Auto-numbering prefixes
NEGATIVE_QTY_ALLOWED_FLAG	VARCHAR2(1)	Y	Item-level override of org rule (default inherit)

Group: Costing & Purchasing

Column	Type	Req	Description
COSTING_ENABLED_FLAG	VARCHAR2(1)	Y	Valued in cost org (default Y)
STANDARD_COST	NUMBER	N	Fallback cost when no item-cost row
PURCHASABLE_FLAG	VARCHAR2(1)	Y	Phase 2 Purchasing
DEFAULT_BUYER	VARCHAR2(150)	N	Phase 2
LIST_PRICE	NUMBER	N	Default purchase price (Phase 2)
MARKET_PRICE	NUMBER	N	Informational
PURCHASING_UOM_CODE	VARCHAR2(10)	N	Default ordering UOM
RECEIPT_ROUTING_OVERRIDE	VARCHAR2(15)	N	Overrides org receiving routing (Phase 2)
PURCHASE_TAX_CODE	VARCHAR2(30)	N	Input tax classification (existing Tax module codes)
INSPECTION_REQUIRED_FLAG	VARCHAR2(1)	Y	Forces INSPECTION routing on receipt (default N)
INVOICE_MATCH_OPTION	VARCHAR2(10)	N	2_WAY / 3_WAY - AP matching rule (Phase 2)
UN_NUMBER / HAZARD_CLASS	VARCHAR2(30)	N	Dangerous-goods references (informational)

Group: Order Management & Invoicing

Column	Type	Req	Description
SALEABLE_FLAG	VARCHAR2(1)	Y	Item may appear on sales orders (Phase 3)
ORDERABLE_FLAG	VARCHAR2(1)	Y	Customer-orderable now vs internal-only (default Y)
SHIPPABLE_FLAG	VARCHAR2(1)	Y	Physically shipped (N for services/fees)
INVOICEABLE_FLAG	VARCHAR2(1)	Y	Generates AR invoice lines (default Y)
ATP_CHECK_FLAG	VARCHAR2(1)	Y	Availability check at order entry (default Y)
RETURNABLE_FLAG	VARCHAR2(1)	Y	Customer returns allowed (default Y)
SELLING_PRICE	NUMBER	N	Default selling price
SALES_UOM_CODE	VARCHAR2(10)	N	Default sales UOM
OVERSHIP_TOLERANCE_PCT / UNDERSHIP_TOLERANCE_PCT	NUMBER	N	Fulfilment tolerances (Phase 3)
SALES_TAX_CODE	VARCHAR2(30)	N	Output tax classification (existing Tax module codes, e.g. UAE VAT 5% / zero-rated / exempt)
TAXABLE_FLAG	VARCHAR2(1)	Y	Line is taxable on sales documents (default Y)

Group: Accounting

Default distribution accounts used when documents for this item are accounted. All are full code combinations picked with AccountSelector; each is org-overridable on the assignment row (5.4), and document-level entry can still override at transaction time.

Column	Type	Req	Description
SALES_ACCOUNT	VARCHAR2(500)	N	Revenue account credited by AR invoicing (Phase 3)
SALES_RETURN_ACCOUNT	VARCHAR2(500)	N	Returns / credit memos (falls back to sales account)
COGS_ACCOUNT	VARCHAR2(500)	N	Cost of goods sold debited on SO_ISSUE (Phase 3)
EXPENSE_ACCOUNT	VARCHAR2(500)	N	Charged when issued to expense / non-asset subinventory
PURCHASE_PRICE_VARIANCE_ACCOUNT	VARCHAR2(500)	N	PO price vs cost variance (Phase 2)
DEFERRED_REVENUE_ACCOUNT	VARCHAR2(500)	N	Reserved for revenue-recognition scenarios

Group: Physical

Column	Type	Req	Description
UNIT_WEIGHT / WEIGHT_UOM_CODE	NUMBER / VARCHAR2(10)	N	Physical
UNIT_VOLUME / VOLUME_UOM_CODE	NUMBER / VARCHAR2(10)	N	Physical
LENGTH / WIDTH / HEIGHT / DIMENSION_UOM_CODE	NUMBER / VARCHAR2(10)	N	Physical

Group: Planning + spare attributes

Column	Type	Req	Description
PLANNING_METHOD	VARCHAR2(15)	Y	NONE / MIN_MAX (default NONE)
MIN_QTY / MAX_QTY	NUMBER	N	Master defaults; org-overridable
FIXED_LOT_MULTIPLIER	NUMBER	N	Suggested order multiple
LEAD_TIME_DAYS	NUMBER	N	Replenishment lead time
SAFETY_STOCK_QTY	NUMBER	N	Org-overridable
ATTRIBUTE1..ATTRIBUTE10	VARCHAR2(240)	N	Flexfield-style spare attributes
Audit columns	-	Y	-

5.2 RR_INV_ITEM_STATUSES

An item status is a template of usage flags; setting the status applies its flag values to the item (each can still be tightened manually). Columns: ITEM_STATUS_CODE PK, NAME, and default values for STOCKABLE / TRANSACTABLE / RESERVABLE / PURCHASABLE / SALEABLE.

Seeded status	Effect
ACTIVE	All flags Y
PROTOTYPE	Stockable/transactable Y, purchasable Y, saleable N
ON_HOLD	Transactable N, reservable N
PHASE_OUT	Purchasable N, saleable Y (sell remaining stock)
OBSOLETE	All usage flags N

5.3 Categories and catalogs

A category set (catalog) is an independent way of classifying items; an item holds at most one category per set. Two sets are seeded: PRODUCT (functional grouping used in screens and reports) and COST (used by costing reports).

Table	Columns
RR_INV_CATEGORY_SETS	CATEGORY_SET_ID PK, SET_NAME unique, DESCRIPTION, MANDATORY_FLAG, STATUS, audit
RR_INV_CATEGORIES	CATEGORY_ID PK, CATEGORY_SET_ID FK, CATEGORY_CODE, CATEGORY_NAME, PARENT_CATEGORY_ID (hierarchy), STATUS, audit; UNIQUE(set, code)
RR_INV_ITEM_CATEGORIES	ITEM_CATEGORY_ID PK, ITEM_ID FK, CATEGORY_SET_ID FK, CATEGORY_ID FK, audit; UNIQUE(item, set)

5.4 RR_INV_ITEM_ORG_ASSIGNMENTS (with org-level overrides)

Column	Type	Req	Description
ASSIGNMENT_ID	NUMBER	Y	PK
ITEM_ID	NUMBER	Y	FK
ORGANIZATION_ID	NUMBER	Y	FK to CHILD org; UNIQUE(item, org)
STATUS	VARCHAR2(10)	Y	ACTIVE / INACTIVE
DEFAULT_SUBINVENTORY	VARCHAR2(30)	N	Item default in this org
DEFAULT_LOCATOR_ID	NUMBER	N	When locator controlled
MIN_QTY / MAX_QTY / SAFETY_STOCK_QTY	NUMBER	N	Org overrides of planning attributes
SOURCE_ORG_ID	NUMBER	N	Replenishment source org (feeds transfer orders)
SALES_ACCOUNT_OVERRIDE	VARCHAR2(500)	N	Org-level revenue account override
COGS_ACCOUNT_OVERRIDE	VARCHAR2(500)	N	Org-level COGS account override
EXPENSE_ACCOUNT_OVERRIDE	VARCHAR2(500)	N	Org-level expense account override
SALES_TAX_CODE_OVERRIDE / PURCHASE_TAX_CODE_OVERRIDE	VARCHAR2(30)	N	Org-level tax classification overrides
Audit columns	-	Y	-

Account and tax resolution order everywhere in the module: document-line entry (if the user typed one) -> org assignment override -> item master value -> organization parameter default. Tax codes reference the existing Re-ERP Tax module's code list.

5.5 RR_INV_ITEM_REVISIONS

Ordered revision labels per item: REVISION_ID PK, ITEM_ID FK, REVISION_CODE (A, B, C...), DESCRIPTION, EFFECTIVE_FROM, audit; UNIQUE(item, code). Transactions optionally capture REVISION_CODE; on-hand is not split by revision in Phase 1.

5.6 Item attachments

RR_INV_ITEM_ATTACHMENTS reuses the proven attachment mechanism (payments/retirements/assets): identity PK, ITEM_ID key column, FILE_NAME, FILE_SIZE, MIME_TYPE, FILE_CONTENT BLOB, DESCRIPTION, STATUS, UPLOADED_BY, UPLOAD_DATE; endpoints inv/items/:itemId/attachments (list, base64 upload, binary download, soft delete); the shared attachments component renders the UI.

6. Lot and Serial Control

Lot and serial control activate per item (flags in 5.1) inside organizations that enable them (4.2). When active, every transaction leg must carry lot/serial detail covering the full quantity, and on-hand is kept per lot.

6.1 RR_INV_LOTS

Column	Type	Req	Description
LOT_ID	NUMBER	Y	PK
ITEM_ID	NUMBER	Y	FK
ORGANIZATION_ID	NUMBER	Y	FK (lots are org-specific)
LOT_NUMBER	VARCHAR2(80)	Y	UNIQUE(item, org, lot number)
ORIGINATION_DATE	DATE	Y	First receipt date
EXPIRATION_DATE	DATE	N	Required when item is expiration controlled
MATERIAL_STATUS_CODE	VARCHAR2(30)	N	Lot-level status (4.9)
SUPPLIER_LOT	VARCHAR2(80)	N	Vendor's lot reference
Audit columns	-	Y	-

6.2 RR_INV_SERIALS

Column	Type	Req	Description
SERIAL_ID	NUMBER	Y	PK
ITEM_ID	NUMBER	Y	FK
SERIAL_NUMBER	VARCHAR2(80)	Y	UNIQUE(item, serial)
CURRENT_ORG_ID	NUMBER	N	Where the unit currently is
CURRENT_SUBINVENTORY / CURRENT_LOCATOR_ID	VARCHAR2/NUMBER	N	Current position
CURRENT_STATUS	VARCHAR2(12)	Y	IN_STOCK / ISSUED / IN_TRANSIT / RETIRED
LOT_ID	NUMBER	N	When both controls active
Audit columns	-	Y	-

6.3 Transaction detail tables

RR_INV_TXN_LOT_DETAILS: DETAIL_ID PK, TRANSACTION_ID FK, LOT_ID FK, QUANTITY - the sum of lot details must equal the leg quantity. RR_INV_TXN_SERIAL_DETAILS: DETAIL_ID PK, TRANSACTION_ID FK, SERIAL_ID FK - one row per unit; the engine moves CURRENT_* fields on the serial master. On-hand rows carry LOT_ID when the item is lot controlled, so availability and expiry reporting work per lot. Issue suggestion follows earliest expiration first for expiration-controlled items, otherwise first-in-first-out by lot origination date.

7. Costing: Cost Organizations, Item Costs, Cost Periods

7.1 RR_INV_COST_ORGANIZATIONS

Column	Type	Req	Description
COST_ORG_ID	NUMBER	Y	PK
COST_ORG_CODE	VARCHAR2(30)	Y	Unique
COST_ORG_NAME	VARCHAR2(240)	Y	-
COST_METHOD	VARCHAR2(15)	Y	AVERAGE / STANDARD
CURRENCY_CODE	VARCHAR2(15)	Y	Costing currency
GL_CALENDAR_ALIGNED_FLAG	VARCHAR2(1)	Y	Y = cost periods mirror GL periods (default Y)
STATUS	VARCHAR2(10)	Y	ACTIVE / INACTIVE
Audit columns	-	Y	-

RR_INV_COST_ORG_ASSIGNMENTS: ASSIGNMENT_ID PK, COST_ORG_ID FK, ORGANIZATION_ID FK UNIQUE (each inventory org belongs to exactly one cost org), audit.

7.2 Cost elements

Unit cost is not a single number: it is the sum of cost elements, so freight and overhead can be tracked and adjusted separately from material value. RR_INV_COST_ELEMENTS is a seeded, extensible lookup: ELEMENT_CODE PK, ELEMENT_NAME, DISPLAY_ORDER, STATUS. Seeded elements: MATERIAL, FREIGHT, HANDLING, OVERHEAD, NON_REC_TAX. Every stored cost and every cost adjustment is expressed per element; the transaction engine always values with the total.

7.3 RR_INV_ITEM_COSTS, cost details and cost history

Column	Type	Req	Description
ITEM_COST_ID	NUMBER	Y	PK
COST_ORG_ID	NUMBER	Y	FK
ITEM_ID	NUMBER	Y	FK; UNIQUE(cost org, item) - the CURRENT cost row
COST_PER_UNIT	NUMBER	Y	Current TOTAL unit cost (sum of elements)
COST_QTY	NUMBER	Y	Quantity basis of the current average (7.5)
COST_SOURCE	VARCHAR2(15)	Y	MANUAL / RECEIPT_AVG / ADJUSTMENT / TRANSFER_IN
LAST_TXN_ID	NUMBER	N	Transaction that last moved the average
Audit columns	-	Y	-

RR_INV_ITEM_COST_DETAILS: DETAIL_ID PK, ITEM_COST_ID FK, ELEMENT_CODE FK, COST_PER_UNIT - one row per element, summing to the parent total. RR_INV_ITEM_COST_HISTORY records one row per change (event RECEIPT / ADJUSTMENT / TRANSFER_IN, transaction id, old average, new average, quantity basis, date, user), so the average is fully auditable through the transaction sequence.

7.4 Perpetual average cost engine (transaction-sequence dependent)

The average moves ONLY on value-bearing inflows; outflows consume at the current average and never move it. Because of this, transaction order matters and the engine processes strictly in creation sequence within a cost organization:

On RECEIPT of qty Q at unit cost C (misc receipt with entered cost, PO receipt in Phase 2, transfer-in from another COST org):

$$\text{new_avg} = (\text{onhand_qty} \times \text{current_avg} + Q \times C) / (\text{onhand_qty} + Q)$$

per element: same formula element-by-element

On ISSUE / transfer-out / count adjustment: value = qty x current_avg
(average unchanged; COST_QTY basis reduced)

Worked example (item FG-1001):

- 1 Receive 100 @ 14.00 -> on-hand 100, avg 14.00
- 2 Issue 40 -> valued at 14.00, avg stays 14.00
- 3 Receive 60 @ 16.50 -> avg = $(60 \times 14.00 + 60 \times 16.50) / 120 = 15.25$
- 4 Issue 30 -> valued at 15.25

Reversing steps 2 and 3 gives a different avg - sequence matters, which is why the engine serializes cost updates per cost org + item (SELECT ... FOR UPDATE on the current cost row).

Negative or zero on-hand receipt edge case: when on-hand <= 0 the new receipt cost becomes the average (no blending with negative quantity).

7.5 Cost adjustments

A cost adjustment revalues on-hand to a new unit cost (total or a single element) without moving quantity. RR_INV_COST_ADJUSTMENTS: ADJUSTMENT_ID PK, COST_ORG_ID, ITEM_ID, ELEMENT_CODE (NULL = total), OLD_COST, NEW_COST, ONHAND_QTY at adjustment time, ADJUSTMENT_VALUE = qty x (new - old), REASON, STATUS, audit. Posting writes a COST_ADJ transaction (zero quantity, value = adjustment value) with accounting Dr Material / Cr Cost-adjustment account when value is positive, reversed when negative, updates the current cost row and writes cost history. Validations: item must have on-hand basis or the adjustment simply sets the go-forward cost; adjustment date must be in an OPEN cost period; approval required above a configurable value threshold.

7.6 Inter-organization transfer costing

Within one cost organization, a transfer moves quantity only - the average is unchanged because value stays in the same cost book (in-transit carries it between the legs). Across DIFFERENT cost organizations, the receiving cost org treats the receipt as an inflow at the SOURCE org's current average (the transfer price), and re-averages exactly as in 7.4; the source org issues at its average. The inter-org receivable/payable lines carry the same value on both sides so the group nets to zero; intercompany markup is a Phase 2+ open decision.

7.7 RR_INV_COST_PERIODS

Cost periods gate transactions and provide the close cycle (checklist below is referenced as 7.7). When GL-aligned, period rows are generated from the existing GL calendar; otherwise they are entered manually.

Column	Type	Req	Description
COST_PERIOD_ID	NUMBER	Y	PK
COST_ORG_ID	NUMBER	Y	FK; UNIQUE(cost org, period name)
PERIOD_NAME	VARCHAR2(30)	Y	e.g. Sep-26 (matches GL period naming)
START_DATE / END_DATE	DATE	Y	-
STATUS	VARCHAR2(15)	Y	NEVER_OPENED / OPEN / PENDING_CLOSE / CLOSED
CLOSED_BY / CLOSED_DATE	VARCHAR2 / DATE	N	-
Audit columns	-	Y	-

Period lifecycle and close checklist

NEVER_OPENED -> OPEN -> PENDING_CLOSE -> CLOSED

Open: only the period after the latest OPEN/CLOSED period can be opened.

Gate: every transaction date must fall in an OPEN cost period of the org's cost organization AND an open GL period.

Close (PENDING_CLOSE -> CLOSED) requires, per cost org:

[1] no unaccounted transactions in the period (ACCOUNTED_FLAG = N count = 0)

[2] no in-transit shipments older than the period end awaiting receipt (warning)

[3] no open physical inventories or unposted count adjustments (blocking)

[4] snapshot written to RR_INV_PERIOD_BALANCES

Reopen: a CLOSED period can be reopened only if the next period is not CLOSED.

7.8 RR_INV_PERIOD_BALANCES

Period-end valuation snapshot per cost org / inventory org / subinventory / item: BALANCE_ID PK, COST_PERIOD_ID FK, ORGANIZATION_ID, SUBINVENTORY_CODE, ITEM_ID, END_QTY, UNIT_COST, END_VALUE, audit. Written at close; the valuation report reads closed periods from here and the open period live from on-hand x current cost.

8. Transaction Engine

8.1 RR_INV_TXN_TYPES (seeded + user-defined)

TXN_TYPE_CODE	Name	Direction	Accounting (Dr / Cr)
MISC_RECEIPT	Miscellaneous Receipt	+	Material / Adjustment (or alias account)
MISC_ISSUE	Miscellaneous Issue	-	Adjustment (or alias) / Material
SUBINV_TRANSFER	Subinventory Transfer	- and + same org	Material(to) / Material(from)
INTERORG_DIRECT	Direct Inter-Org Transfer	- source, + destination	Section 10
INTERORG_SHIP	In-Transit Shipment	- source, + transit	Section 10
INTERORG_RECEIVE	In-Transit Receipt	- transit, + destination	Section 10
CYCLE_COUNT_ADJ	Cycle Count Adjustment	+/-	Material / Cycle count adj account
PHYS_INV_ADJ	Physical Inventory Adjustment	+/-	Material / Physical inv adj account
PO_RECEIPT	PO Receipt	+	Reserved - Phase 2
RETURN_TO_VENDOR	Return To Vendor	-	Reserved - Phase 2
SO_ISSUE	Sales Order Issue	-	Reserved - Phase 3
COST_ADJ	Cost Adjustment	0 qty	Dr/Cr Material vs Cost-adjustment account (7.5) - ACTIVE in Phase 1g

Transaction types are a maintainable setup, not a fixed list. Each type row carries: TXN_TYPE_CODE PK, NAME, SYSTEM_ACTION (RECEIPT / ISSUE / TRANSFER - fixed behaviors the engine implements), DEFAULT_OFFSET_ACCOUNT, DEFAULT_REASON_CODE, USER_DEFINED_FLAG, STATUS. Users create their own types on the seeded system actions - e.g. PRODUCTION-SCRAP (ISSUE, offset = scrap expense), FREE-SAMPLE-OUT (ISSUE, offset = marketing expense), CUSTOMER-RETURN-IN (RECEIPT) - and they appear in Create Transaction with their default offset account and reason prefilled. Seeded types cannot be deleted; user types with history can only be deactivated.

8.2 RR_INV_TRANSACTIONS

Column	Type	Req	Description
TRANSACTION_ID	NUMBER	Y	PK
ORGANIZATION_ID	NUMBER	Y	Org of this leg
TXN_TYPE_CODE	VARCHAR2(30)	Y	FK
ITEM_ID	NUMBER	Y	FK
REVISION_CODE	VARCHAR2(10)	N	Optional capture
SUBINVENTORY_CODE	VARCHAR2(30)	Y	-
LOCATOR_ID	NUMBER	N	When controlled
QUANTITY	NUMBER	Y	Signed; stored in primary UOM
ENTERED_QTY / ENTERED_UOM_CODE	NUMBER / VARCHAR2(10)	Y	As typed by the user
UNIT_COST	NUMBER	Y	Applied cost
TXN_VALUE	NUMBER	Y	QUANTITY x UNIT_COST (signed)
TRANSACTION_DATE	DATE	Y	Accounting date

Column	Type	Req	Description
REASON_CODE	VARCHAR2(30)	N	FK to reasons
ACCOUNT_ALIAS_ID	NUMBER	N	FK; replaces adjustment account on misc types
TRANSFER_GROUP_ID	NUMBER	N	Pairs subinventory-transfer legs
TRANSFER_ORG_ID / TRANSFER_SUBINVENTORY	NUMBER / VARCHAR2(30)	N	Counterpart on transfers
SHIPMENT_ID	NUMBER	N	FK inter-org shipments
TRANSFER_ORDER_LINE_ID	NUMBER	N	FK when executing a transfer order
COUNT_ENTRY_ID / PHYS_TAG_ID	NUMBER	N	Backlink from count adjustments
SOURCE_TYPE	VARCHAR2(30)	Y	INV now; PO / OM / COST reserved
SOURCE_REFERENCE	VARCHAR2(80)	N	Document reference
ACCOUNTED_FLAG	VARCHAR2(1)	Y	N until posted
JE_BATCH_ID	NUMBER	N	GL batch
Audit columns	-	Y	-

8.3 Processing algorithm

```

POST inv/transactions/create
1 Org access check (2.1); org and item ACTIVE; assignment ACTIVE in org
2 Item status flags allow the action (transactable; reservable for step 6)
3 Subinventory / locator exist, ACTIVE; material status of source and
  destination allows the action (4.9)
4 Date inside an OPEN cost period (7.7) and an open GL period
5 UOM: convert entered qty to primary using item conversion else standard
6 Availability on negative legs: on-hand minus active reservations must
  cover the qty, unless negative inventory is allowed (org + item rule);
  reserved stock can only be issued by its own demand reference
7 Lot/serial: details required when controlled; expiry validated; serial
  state transitions applied
8 Cost: item cost in the org's cost org, else item standard cost, else 409
9 Insert transaction leg(s); MERGE on-hand (per lot when controlled);
  update lot/serial masters - one database transaction
10 Build Dr/Cr rows in RR_INV_TXN_ACCOUNTING from the type rules
11 COMMIT; return ids, applied cost, new on-hand

```

8.4 Accounting hand-off

The Create Accounting action groups unposted RR_INV_TXN_ACCOUNTING rows by organization and date into one journal batch through the existing GL journal interface, then writes back JE_BATCH_ID and ACCOUNTED_FLAG = Y. Journal lines carry REFERENCE2 = TRANSACTION_ID and REFERENCE5 = 'INV_TRANSACTIONS', the same convention AR/AP/Cash use, so Check Accounting and the journal drill link back to inventory automatically. RR_INV_TXN_ACCOUNTING columns: ACCT_LINE_ID PK, TRANSACTION_ID FK, LINE_TYPE DR/CR, ACCOUNT_COMBINATION, AMOUNT, CURRENCY_CODE, POSTED_FLAG, JE_BATCH_ID.

9. Reservations and Availability

A reservation ring-fences quantity for a demand source so other transactions cannot consume it. Available-to-transact (ATT) everywhere in the module is: on-hand minus active reservations at the same level.

9.1 RR_INV_RESERVATIONS

Column	Type	Req	Description
RESERVATION_ID	NUMBER	Y	PK
ORGANIZATION_ID	NUMBER	Y	FK
ITEM_ID	NUMBER	Y	FK
SUBINVENTORY_CODE	VARCHAR2(30)	N	NULL = org-level reservation
LOCATOR_ID / LOT_ID	NUMBER	N	Optional detail level
RESERVED_QTY	NUMBER	Y	In primary UOM
FULFILLED_QTY	NUMBER	Y	Consumed so far (default 0)
DEMAND_SOURCE_TYPE	VARCHAR2(30)	Y	MANUAL now; SALES_ORDER / TRANSFER_ORDER / WORK_ORDER reserved
DEMAND_REFERENCE	VARCHAR2(80)	N	Document / free reference
NEED_BY_DATE	DATE	N	-
STATUS	VARCHAR2(12)	Y	ACTIVE / FULFILLED / CANCELLED
Audit columns	-	Y	-

9.2 Rules

- Creating a reservation checks ATT at the requested level and rejects shortfalls
- Issues quoting the same demand reference consume the reservation (FULFILLED_QTY up; auto-FULFILLED when complete); other issues see only unreserved stock
- Transfers and count adjustments that would drop on-hand below reserved quantity are rejected with a clear message listing the blocking reservations
- Reservations page: create / view / cancel, with ATT preview per item and level; Phase 3 creates reservations automatically from sales orders

10. Inter-Org Transfers and Transfer Orders

10.1 Direct and in-transit movements

DIRECT one step: - source subinv, + destination subinv
Source: Dr Inter-org Receivable Cr Material
Destination: Dr Material Cr Inter-org Payable

INTRANSIT step 1 SHIP: - source subinv, + transit value
Source: Dr In-transit account Cr Material
step 2 RECEIVE: - transit, + destination subinv
Destination: Dr Material Cr In-transit account

Transfer price = item cost in the SOURCE org's cost organization; a destination in a different cost org books the same value (no markup in Phase 1; intercompany invoicing is a Phase 2+ decision).

RR_INV_INTERORG_SHIPMENTS: SHIPMENT_ID PK, SHIPMENT_NUMBER (from shipping parameters), FROM_ORG_ID, TO_ORG_ID, TRANSFER_MODE DIRECT/INTRANSIT, STATUS SHIPPED/RECEIVED, SHIP_DATE, RECEIVE_DATE, SHIP_METHOD, NOTES, audit. Lines are the transaction rows carrying SHIPMENT_ID. In-transit quantity appears on availability screens as an IN-TRANSIT bucket per destination org.

10.2 Transfer orders (requested transfers with a lifecycle)

A transfer order is the documented request to move stock between organizations, with approval and partial fulfilment, executed through the shipment flow above.

Table	Columns
RR_INV_TRANSFER_ORDERS	TRANSFER_ORDER_ID PK, TO_NUMBER unique (TO-000001), FROM_ORG_ID, TO_ORG_ID, TRANSFER_MODE, STATUS DRAFT/REQUESTED/APPROVED/IN_PROGRESS/COMPLETED/CANCELLED, REQUESTED_BY, APPROVED_BY/DATE, NEED_BY_DATE, NOTES, audit
RR_INV_TRANSFER_ORDER_LINES	LINE_ID PK, TRANSFER_ORDER_ID FK, LINE_NUM, ITEM_ID, REQUESTED_QTY, SHIPPED_QTY, RECEIVED_QTY, UOM_CODE, SOURCE_SUBINVENTORY, DEST_SUBINVENTORY, STATUS OPEN/PARTIAL/CLOSED/CANCELLED, audit

DRAFT -> REQUESTED -> APPROVED -> IN_PROGRESS -> COMPLETED

\-> CANCELLED (until first shipment)

Ship against an APPROVED order: creates a shipment whose lines reference the TO lines and roll SHIPPED_QTY / RECEIVED_QTY up; the order completes when all lines are fully received. Min-max suggestions (5.4 SOURCE_ORG_ID) generate DRAFT transfer orders in a later phase.

11. Counting: Cycle Count, Physical Inventory, ABC

11.1 ABC analysis

An ABC compile ranks items in an organization by a chosen criterion and assigns classes used to drive cycle-count frequency.

Table	Columns
RR_INV_ABC_COMPILES	COMPILE_ID PK, ORGANIZATION_ID, COMPILE_NAME, CRITERION ONHAND_VALUE / ISSUE_VALUE_12M / ISSUE_QTY_12M, COMPILE_DATE, STATUS, audit
RR_INV_ABC_CLASSES	CLASS_ID PK, COMPILE_ID FK, CLASS_CODE A/B/C..., PCT_OF_VALUE, description
RR_INV_ABC_ASSIGNMENTS	ASSIGNMENT_ID PK, COMPILE_ID FK, ITEM_ID, CLASS_CODE, RANK, VALUE_BASIS, audit

11.2 Cycle counting

Users create any number of cycle count definitions per organization (name, scope by ABC class / category / subinventory / manual item list, counts per year per class, tolerances, approval rule). From a definition, entries are generated on schedule or on demand, counted, and approved into adjustment transactions. A definition with history is deactivated, never deleted.

Table	Columns
RR_INV_CYCLE_COUNTS	CYCLE_COUNT_ID PK, ORGANIZATION_ID, COUNT_NAME, ABC_COMPILE_ID (optional), COUNTS_PER_YEAR per class (A/B/C columns), APPROVAL_REQUIRED_FLAG, QTY_TOLERANCE_PCT, VALUE_TOLERANCE, STATUS, audit
RR_INV_CYCLE_COUNT_ITEMS	CC_ITEM_ID PK, CYCLE_COUNT_ID FK, ITEM_ID, CLASS_CODE, LAST_COUNT_DATE, NEXT_DUE_DATE
RR_INV_CYCLE_COUNT_ENTRIES	ENTRY_ID PK, CYCLE_COUNT_ID FK, ITEM_ID, SUBINVENTORY_CODE, LOCATOR_ID, LOT_ID, SNAPSHOT_QTY, COUNTED_QTY, COUNTED_BY/DATE, VARIANCE_QTY/VALUE, STATUS PENDING/COUNTED/APPROVED/REJECTED/RECOUNT, ADJ_TRANSACTION_ID, audit

Generate entries (due items) -> print/count -> enter counted qty
variance within tolerance and no approval required -> auto-approve
else supervisor approves / requests recount
Approve -> CYCLE_COUNT_ADJ transaction for the variance (links back via ADJ_TRANSACTION_ID) -> on-hand corrected, accounting per 8.1

11.3 Physical inventory

Users create any number of physical inventories (name, organization, scope: all subinventories or a chosen list, tolerance, approval rule). Each runs the wizard below independently, so several warehouses can count in parallel.

Table	Columns
RR_INV_PHYSICAL_INVENTORIES	PHYSICAL_INV_ID PK, ORGANIZATION_ID, PHYS_INV_NAME, SNAPSHOT_DATE, FREEZE_FLAG (blocks transactions in scope while Y), SCOPE all/subinventory list, APPROVAL_TOLERANCE_PCT, STATUS DRAFT/SHOT/COUNTING/ADJUSTING/COMPLETED, audit
RR_INV_PHYS_INV_SNAPSHOT	SNAP_ID PK, PHYSICAL_INV_ID FK, ITEM_ID, SUBINVENTORY_CODE, LOCATOR_ID, LOT_ID, SNAPSHOT_QTY, UNIT_COST
RR_INV_PHYS_INV_TAGS	TAG_ID PK, PHYSICAL_INV_ID FK, TAG_NUMBER, ITEM_ID, SUBINVENTORY_CODE, LOCATOR_ID, LOT_ID, COUNTED_QTY, COUNTED_BY/DATE, STATUS OPEN/ENTERED/VOID, audit

Create -> take SNAPSHOT (freeze scope) -> generate/print tags -> enter counts
-> review variances (tag vs snapshot) -> approve -> PHYS_INV_ADJ transactions
-> unfreeze -> COMPLETED. Cost periods cannot close with an open physical inventory in scope (7.3).

12. Inquiry and Reporting

- On-Hand Inquiry: org / subinventory / locator / item / lot levels, ATT column (on-hand minus reservations), in-transit bucket, expiring-lots filter, Excel export, drill to transaction history
- Transaction History: filters org, item, type, date range, reason, source; drill to accounting lines and to the GL journal (Full Journal preview component)
- Valuation report: open period live (on-hand x current cost), closed periods from RR_INV_PERIOD_BALANCES, grouped by cost org / org / subinventory / category
- Reservations report: active reservations vs on-hand by item
- Count accuracy report: cycle count hit rate and net adjustment value by period
- All inquiries run through the guarded SQL gateway (appUser INV_MANAGEMENT) or dedicated GET endpoints; every screen keeps the API-transparency button

13. Service Architecture - Gateway + Packages (no per-feature endpoints)

The module deliberately does NOT create one ORDS endpoint per feature. All business logic lives in PL/SQL packages, and the frontend reaches the database through just two generic gateways (plus the standard attachment endpoints for BLOBs). This is the same execution style the AI assistant and Check Accounting already use, extended with a safe write path.

13.1 Read path - POST ai/executequery (existing)

Every inquiry screen issues plain SELECT statements through the existing guarded SELECT-only gateway with appUser INV_MANAGEMENT. Reads never call procedures; they select from the RR_INV_* tables and from one list view per screen so column shapes stay stable when tables evolve:

View	Feeds
RR_INV_V_ORGS	Inventory Organizations grid (org + BU + parameter highlights)
RR_INV_V_SUBINVENTORIES	Subinventories tab (+ locator counts)
RR_INV_V_ITEMS	Item Master grid (attribute groups flattened, category, assignments list)
RR_INV_V_ITEM_COSTS	Item Costs tab (avg, basis, element breakdown)
RR_INV_V_COST_HISTORY	Cost history drill
RR_INV_V_ONHAND	On-Hand & Availability (on-hand, reserved, ATT, in-transit)
RR_INV_V_TRANSACTIONS	Transaction History (+ accounting join)
RR_INV_V_RESERVATIONS	Reservations page
RR_INV_V_SHIPMENTS / RR_INV_V_TRANSFER_ORDERS	Transfers pages
RR_INV_V_COUNT_ENTRIES / RR_INV_V_PHYS_TAGS	Counting pages
RR_INV_V_COST_PERIODS / RR_INV_V_PERIOD_BALANCES	Cost Periods / Valuation

13.2 Write path - POST inv/execute (ONE new endpoint)

A single dispatcher endpoint executes whitelisted package procedures. The client sends the procedure name and NAMED parameters as JSON; the handler binds them - parameter values are never concatenated into SQL, which keeps the gateway injection-proof while still being 'direct to the database'.

```
Request:
POST {base}/inv/execute
{ "procedure": "RR_INV_ORG_PKG.CREATE_ORG",
  "appUser": "KHALID",
  "params": { "p_org_code": "W3", "p_org_name": "Sharjah Warehouse",
              "p_org_type": "CHILD", "p_master_org_code": "MST",
              "p_business_unit": "BUIMERC CORP_DIFC_INVST" } }
```

Handler logic (plsql/block):

```

1  parse JSON; look the procedure up in RR_INV_PROC_REGISTRY
   (proc name, expected params, enabled flag) - unknown/disabled -> 403
2  build 'BEGIN <proc>(p_org_code=>:1, ..., p_user=>:u,
      p_id=>:id, p_status=>:st, p_message=>:msg); END;'
   binding every value by name from the JSON (native dynamic SQL)
3  execute; the PACKAGE runs all validations (section 17) and business
   logic (engine 8.3, cost engine 7.4 ...) and raises/returns cleanly
4  write RR_INV_EXEC_LOG (user, proc, params CLOB, status, elapsed ms)
5  respond { "success": true, "id": 5000000123, "message": "..." }
   or      { "success": false, "error": "<p_message>" } with 400/403/409

```

Every procedure follows one signature convention:

```

IN parameters  p_xxx  (strings/numbers/dates as VARCHAR2 ISO)
IN             p_user (audit stamp; org access is checked inside)
OUT           p_id NUMBER, p_status VARCHAR2 ('S'/'E'), p_message VARCHAR2

```

13.3 Package specifications - every procedure and function, with its validations

Convention for all procedures: IN parameters p_xxx, IN p_user (audit + org-access check), OUT p_id NUMBER, OUT p_status VARCHAR2 ('S'/'E'), OUT p_message VARCHAR2. Functions are pure reads used by the engine and other packages. Validation references (17.x, R0-R9, I0-I9) point to the rule catalogs; rules named here are the procedure-specific ones on top.

13.3.1 RR_INV_ORG_PKG - organization and warehouse structure

Why: everything physical hangs off the organization; this package owns the hierarchy, its parameter sets, subinventories/locators and who may transact where.

Procedure / Function	Validations enforced
CREATE_ORG (code, name, type, master code, BU, location)	code required/uppercase/unique; type MASTER or CHILD; CHILD requires existing master + BU from gl/businessunits (LE copied); one MASTER per company (17.1)
UPDATE_ORG (id, name, location, status)	org exists; deactivate blocked while on-hand, open shipments/TOs/counts exist (17.1)
SET_PARAMETERS (org, all 4.2 fields)	org exists; UOM exists; every account a valid combination; flags Y/N; locator control value in list
SET_RECEIVING_PARAMS / SET_SHIPPING_PARAMS	routing/numbering values in list; tolerances >= 0; counters only increase (17.1)
CREATE_SUBINV (org, code, type, flags, accounts)	unique code per org; QUANTITY_TRACKED=N forces ASSET=N; material status exists (17.2)
UPDATE_SUBINV	deactivate blocked while on-hand or open count entries (17.2)
CREATE_LOCATOR (subinv, code, ...)	subinventory is locator-controlled; code unique within it (17.2)
GRANT_ORG_ACCESS / REVOKE_ORG_ACCESS (user, org ALL)	user required; org exists; duplicate grant idempotent; revoke leaves audit
F GET_ORG_ID(code), F HAS_ACCESS(user, org)	lookups used by every other package (HAS_ACCESS implements 2.1)

13.3.2 RR_INV_ITEM_PKG - item master

Why: single owner of item definitions and their organization footprint; no other package writes item rows.

Procedure / Function	Validations enforced
CREATE_ITEM (all 5.1 attribute groups; categories JSON)	item number required/unique/trimmed; UOM + status + tax codes + categories exist; flag consistency: expiration control requires lot control, shelf life requires expiration control; accounts valid combinations; mandatory category sets satisfied at ACTIVE status (17.3)
UPDATE_ITEM	conflicts with existing stock blocked: cannot disable lot/serial control or stockable while on-hand exists (17.3)
SET_ITEM_STATUS (item, status)	status exists; applies the template flags (5.2); OBSOLETE requires zero on-hand or explicit force with reason
ASSIGN_TO_ORG (item, org, defaults, overrides)	CHILD org; item ACTIVE; duplicate assignment rejected; default subinv/locator exist in that org; override accounts/tax codes valid (17.3)
UPDATE_ASSIGNMENT	deactivate blocked while on-hand or active reservations in that org (17.3)
SET_CATEGORY (item, set, category)	set + category exist and ACTIVE; one category per set (17.3)
CREATE_REVISION (item, code, effective)	code unique per item, ordered; effective date valid
F GET_ITEM_ID(number), F ITEM_FLAG(item, flag)	lookups for the engine

13.3.3 RR_INV_LOOKUP_PKG - shared setup lookups

Why: one home for every small controlled list, so screens and packages validate against a single source.

Procedure	Validations enforced
CREATE_TXN_TYPE (code, name, action, offset acct)	code unique; action in RECEIPT/ISSUE/TRANSFER; offset account valid; seeded types protected (8.1)
DEACTIVATE_TXN_TYPE	seeded types refuse; user types with history deactivate only (8.1)
CREATE_REASON / CREATE_SHIP_METHOD / CREATE_MATERIAL_STATUS	code unique; material status flags Y/N complete (4.8-4.10)
CREATE_ACCOUNT_ALIAS (org, code, account)	unique per org; account a valid combination (4.8)
CREATE_UOM / SET_ITEM_UOM_CONVERSION	class in list; one base per class; conversion > 0 (17.6 gate 12 feeds off this)
CREATE_CATEGORY_SET / CREATE_CATEGORY	set name unique; category unique within set; parent in same set (17.3)

13.3.4 RR_INV_TXN_PKG - the transaction engine

Why: the ONLY writer of transactions, on-hand, lot/serial state and accounting rows - every movement in the module funnels through CREATE_TXN so the rules can never be bypassed.

Procedure / Function	Validations enforced
CREATE_TXN (type, org, item, subinv, locator, qty, uom, unit cost, date, reason, alias, demand ref, lots JSON, serials JSON, source refs)	the FULL stack: common gates 17.6 items 1-16, receiving list R1-R9 or issuing list I1-I9 by direction, transfer rules for two-leg calls; atomic write set + post-conditions (Dr=Cr, on-hand reconciliation) before COMMIT (8.3)
CREATE_ACCOUNTING (org, date range)	only unposted rows; every entry balances; GL period open; writes one batch via the journal interface; stamps JE_BATCH_ID + ACCOUNTED_FLAG (8.4)
F PREVIEW_ACCOUNTING(...) RETURN CLOB	same account resolution as CREATE_TXN with zero writes - powers the UI preview
F GET_ONHAND(org,item,subinv,locator,lot), F GET_ATT(...)	balance reads the UI and other packages use (I3 definition)

13.3.5 RR_INV_COST_PKG - costing

Why: owns the perpetual average, elements, adjustments and the period close - the only package allowed to move a cost.

Procedure / Function	Validations enforced
CREATE_COST_ORG / ASSIGN_ORG_TO_COST_ORG	code unique; an inventory org can belong to only one cost org (17.5)
SET_ITEM_COST (cost org, item, elements)	values >= 0; elements sum to total; date in OPEN period (17.5)
ADJUST_COST (cost org, item, element TOTAL, new cost, reason)	element exists; new >= 0; OPEN period; approval above threshold; posts COST_ADJ + history + register (7.5)
OPEN_PERIOD / CLOSE_PERIOD / REOPEN_PERIOD	open only the next period; close checklist: zero unaccounted txns, no open PI, in-transit warning, then snapshot; reopen only while next not CLOSED (7.7)
RECEIPT_REAVERAGE (internal, engine-only)	locked cost row (FOR UPDATE); formula 7.4; zero/negative basis edge case; history row
F GET_ITEM_COST(cost org, item)	current average else standard cost else error signal (R1/I7)

13.3.6 RR_INV_TRANSFER_PKG - transfer orders and shipments

Why: wraps the engine for two-org movements and owns the TO lifecycle and shipment documents.

Procedure	Validations enforced
CREATE_TRANSFER_ORDER (from, to, mode, lines)	orgs differ and ACTIVE; every line item assigned+ACTIVE in BOTH orgs; qty > 0 (17.8)
APPROVE_TO / CANCEL_TO	lifecycle transitions only forward; cancel only before first shipment (17.8)
SHIP (TO id ad-hoc params)	APPROVED TO or valid ad-hoc; required inter-org / in-transit accounts present on both orgs; ship qty <= remaining; source legs via CREATE_TXN (inherits I-list); shipment doc + carried cost written (17.8, 10)
RECEIVE_SHIPMENT (shipment, dest subinvs)	status SHIPPED; receiver has dest-org access; qty <= shipped-not-received; dest legs via CREATE_TXN (inherits R-list); cross-cost-org re-average at carried cost; TO roll-up + COMPLETED (17.8, 7.6)

13.3.7 RR_INV_RES_PKG - reservations

Why: availability protection is its own concern; the engine only consults it.

Procedure	Validations enforced
CREATE_RESERVATION (org, item, level, qty, demand)	item RESERVABLE; qty <= ATT at the requested level; level fields only where the matching control applies (17.7)
CANCEL_RESERVATION	ACTIVE only; fulfilled quantity never reversed (17.7)

13.3.8 RR_INV_COUNT_PKG - counting and ABC

Why: count documents and their approvals; the resulting corrections still post through the engine.

Procedure	Validations enforced
CREATE_CYCLE_DEF (org, scope, frequencies, tolerances)	org exists; scope valid; tolerances >= 0 (11.2)
GENERATE_ENTRIES (def)	due items only; one open entry per item+location; snapshot = current on-hand (17.9)
ENTER_COUNT (entry, qty, counter)	entry PENDING; qty >= 0 (17.9)
APPROVE_ENTRY (entry, approver)	within tolerance auto-approves; above tolerance approver required and must differ from counter; posts CYCLE_COUNT_ADJ via CREATE_TXN (17.9)
REQUEST_RECOUNT	resets to PENDING keeping history
CREATE_PHYS_INV / TAKE_SNAPSHOT / ENTER_TAG / APPROVE_PHYS_INV / COMPLETE_PHYS_INV	scope valid; snapshot freezes scope (engine gate 15); tags unique, VOID needs reason; completion requires all tags ENTERED or VOID; adjustments via CREATE_TXN; unfreeze on completion (17.9)
CREATE_ABC_COMPILE / RUN_COMPILE / SET_CLASSES	criterion data exists; classes cover 100% of value (17.9)

13.4 Frontend service layer

```
// src/services/inv.service.ts - the ONLY two calls the module uses
export const invQuery = (sql: string) =>
  fetch(base + '/ai/executequery', { method: 'POST',
    body: JSON.stringify({ sql, maxRows: 2000, appUser: 'INV_MANAGEMENT' }) });

export const invExec = (procedure: string, params: Record<string, any>) =>
  fetch(base + '/inv/execute', { method: 'POST',
    body: JSON.stringify({ procedure, params, appUser: currentUser }) });

// examples
invQuery("SELECT * FROM rr_inv_v_orgs ORDER BY organization_code");
invExec('RR_INV_ORG_PKG.CREATE_ORG', { p_org_code:'W3', p_org_name:'Sharjah WH',
  p_org_type:'CHILD', p_master_org_code:'MST', p_business_unit:'...' });
invExec('RR_INV_TXN_PKG.CREATE_TXN', { p_txn_type:'MISC_RECEIPT', p_org_code:'W1',
  p_item_number:'FG-1001', p_subinventory:'STORES', p_quantity:100,
  p_unit_cost:14, p_txn_date:'2026-09-21',
  p_lots_json:'[{"lot":"L-2026-001","qty":60}, {"lot":"L-2026-002","qty":40}]' });
```

13.5 Security and operational rules

- RR_INV_PROC_REGISTRY whitelists every callable procedure with its expected parameter names; anything not registered (or disabled) is refused with 403 - the dispatcher can never be steered to other packages or raw DML
- All parameter values are BOUND, never concatenated - the write gateway is immune to SQL injection even though development feels like 'direct SQL'
- The read gateway stays SELECT-only; DML through ai/executequery remains impossible
- Org access (2.1) and every section-17 validation are enforced INSIDE the packages, so the rules hold no matter which client calls
- RR_INV_EXEC_LOG records every call (user, procedure, params, status, elapsed) - the screens' API-transparency dialog shows the exact procedure call and response
- Attachments keep the standard dedicated endpoints (BLOB upload/download does not fit the JSON gateway) - same mechanism as payments/assets
- Benefit: adding a feature = adding a procedure + registry row; no ORDS template work, none of the template-collision failure modes

13.6 Trade-offs acknowledged

- One endpoint means coarse HTTP semantics: errors are expressed in p_status/p_message, not per-endpoint HTTP codes (the dispatcher still maps E->400/403/409 classes)
- The registry must be maintained - a new procedure that is not registered simply cannot be called (fails safe)
- Large result sets still flow through ai/executequery's maxRows cap; paging is done in SQL (OFFSET/FETCH)

14. Navigation and Screen Designs

14.0 Module menu (menuitems.ts entries)

A new Inventory module (module code INV, color slot after CASH) is added to the global menu and to the home tiles. The full menu tree, grouped the way it will appear in the menu search and on the module landing page:

Group	Menu entry	Path
Setup	Inventory Organizations (workspace tabs: Parameters, Receiving, Shipping, Subinventories & Locators, Access)	/inv/organizations
Setup	Cost Organizations (tabs: Assignments, Item Costs, Cost Periods, Period Balances)	/inv/cost-organizations
Setup	Units of Measure	/inv/setup/uoms
Setup	Item Statuses	/inv/setup/item-statuses
Setup	Category Sets & Categories	/inv/setup/categories
Setup	Transaction Reasons	/inv/setup/reasons
Setup	Account Aliases	/inv/setup/account-aliases
Setup	Material Statuses	/inv/setup/material-statuses
Setup	Ship Methods	/inv/setup/ship-methods
Setup	Organization Access	/inv/setup/org-access
Items	Item Master (workspace tabs per section 5)	/inv/items
Items	Lots & Serials	/inv/lots-serials
Transactions	Create Transaction	/inv/transactions/create
Transactions	Transaction History	/inv/transactions
Transactions	Create Accounting	/inv/transactions/accounting
Transfers	Transfer Orders	/inv/transfer-orders
Transfers	Inter-Org Shipments	/inv/shipments
Counting	Cycle Counting	/inv/cycle-counting
Counting	Physical Inventory	/inv/physical-inventory
Counting	ABC Analysis	/inv/abc-analysis
Inquiry	On-Hand & Availability	/inv/onhand
Inquiry	Reservations	/inv/reservations
Inquiry	Valuation Report	/inv/valuation

14.0.1 Screens overview

All pages use the standard Re-ERP frame (breadcrumbs, module-colored title block, search cards, sortable grids with grid search, closable-tab workspaces, API-transparency buttons).

Page	Content
Inventory Organizations	Org grid; org workspace tabs: Parameters, Receiving, Shipping, Subinventories (+ locators drill), Access
Item Master	Item search grid; item workspace tabs: Main, Inventory, Costing & Purchasing, Sales & Physical, Planning, Categories, Revisions, Organizations (assignments with overrides), Attachments
Lots & Serials	Search by item/lot/serial; status and expiry views
Create Transaction	Type-driven form with lot/serial capture grid, reason, account alias, accounting preview

Page	Content
Transaction History	Filtered grid, drill to accounting and Full Journal
On-Hand & Availability	Org/subinv/locator/item/lot levels, ATT and in-transit buckets, expiring lots
Reservations	ATT preview, create/cancel, demand references
Transfer Orders	Lifecycle grid, approve/ship/receive actions, line fulfilment status
Inter-Org Shipments	Shipment grid with Receive action (in-transit)
Cycle Counting	Definitions, due-entry generation, count entry sheet, approval queue
Physical Inventory	Wizard: snapshot -> tags -> counts -> variance review -> adjust
ABC Analysis	Compile runs, class boundaries, assignment review
Cost Organizations	Cost org grid; tabs: Assignments, Item Costs, Cost Periods (open/close with checklist), Period Balances
Setup lookups	UOMs, reasons, account aliases, material statuses, ship methods, item statuses, category sets

14.1 Key wireframes

Item Master - item workspace

```
+-----+
| FG-1001 Steel Bracket 40mm [ACTIVE] [API] [Save] [Close] |
| [Main | Inventory | Costing&Purch | Order Mgmt | Accounting | Physical | |
| Planning | Categories | Revisions | Organizations | Attachments] |
| Inventory tab: Stockable [Y] Transactable [Y] Reservable [Y] |
| Lot control [Y] Expiration control [Y] Shelf life [365] days |
| Serial control [N] Lot prefix [L-] |
| Organizations tab: |
| Org | Status | Default Subinv | Min | Max | Safety | Source Org | [Assign] |
| W1 | ACTIVE | STORES | 50 | 500 | 20 | - | |
+-----+
```

Create Transaction (lot controlled)

```
+-----+
| Type [Misc Receipt v] Org [W1 v] Date [19-Sep-2026] Reason [OPENING v] |
| Item [FG-1001] Subinv [STORES v] Qty [100] UOM [EA v] Cost 14.00 (ro) |
| Lots: | Lot Number | Qty | Expiry | [+ Add lot] remaining: 0 |
| | L-2026-001 | 60 | 19-Sep-27 | |
| | L-2026-002 | 40 | 01-Oct-27 | |
| | [ Preview accounting ] [ Create ] |
+-----+
```

Cost Periods tab

```
+-----+
| Cost Org C0-01 (AVERAGE, AED) |
| Period | Start | End | Status | Actions |
| Aug-26 | 01-Aug-26 | 31-Aug-26 | CLOSED | [Balances] |
| Sep-26 | 01-Sep-26 | 30-Sep-26 | OPEN | [Close...] |
| Oct-26 | 01-Oct-26 | 31-Oct-26 | NEVER_OPENED | [Open] |
| Close dialog shows the checklist live: unaccounted txns 0 [ok], |
| open physical inventories 0 [ok], in-transit warnings 1 [!] -> [Close] |
+-----+
```

Physical Inventory wizard

```
+-----+
| PI-2026-01 Org W1 Status: COUNTING |
| [1 Define] -> [2 Snapshot] -> [3 Tags] -> [4 Counts] -> [5 Variances] -> |
| [6 Adjust & Complete] |
| Variances: Item | Subinv | Snapshot | Counted | Var Qty | Var Value | ok? |
| FG-1001 | STORES | 100 | 97 | -3 | -42.00 | [x] |
| [Approve within tolerance] [Approve selected] [Recount] |
+-----+
```

15. Numbering, Seeding, Conventions

- RR_INV_SEQ starts at 5000000001 and feeds every PK
- Document numbers: receipts and shipments from the per-org counters; transfer orders TO-000001; physical inventories PI-YYYY-NN; tags sequential per physical inventory
- Seeded data: 12 UOMs, transaction types (8.1), item statuses (5.2), material statuses (4.9), reasons (OPENING, DAMAGE, EXPIRY, SAMPLE, CORRECTION, DONATION), category sets PRODUCT and COST, ship methods (ROAD, SEA, AIR, COURIER)
- ORDS rules carried from hard-won lessons: distinct literal paths, spaced "success": literals, ORDS.DEFINE_PARAMETER for every GET parameter, EXECUTE IMMEDIATE for optional tables, one script per feature numbered database/inv/2xx_*.sql, verification query in every script footer
- GL references: REFERENCE2 = TRANSACTION_ID, REFERENCE5 = 'INV_TRANSACTIONS'

16. Delivery Roadmap

Phase	Content	Depends on
1a Setup	Orgs, parameters, subinventories/locators, UOMs, lookups, org access, cost orgs	-
1b Items	Full item master, statuses, categories, revisions, attachments, assignments	1a
1c Engine	Transactions, on-hand, accounting hand-off, transaction history, on-hand inquiry	1b
1d Lots/serials	Lot & serial masters, capture in transactions, per-lot on-hand	1c
1e Transfers	Direct + in-transit shipments, transfer orders	1c
1f Reservations	Reservations and ATT everywhere	1c
1g Costing ops	Cost periods, period close with snapshot, valuation report	1c
1h Counting	Cycle counting, physical inventory, ABC analysis	1c (1d for lot-level counts)
2 Purchasing	Requisitions, POs, PO receiving into the engine (PO_RECEIPT), returns	1c
3 Order Mgmt	Sales orders, auto-reservations, pick/ship (SO_ISSUE)	1f
4 Cost Acctg	Perpetual average, cost adjustments (COST_ADJ), variance and history reports	1g

Each sub-phase ships as one database script plus its pages and is deployable independently, the delivery style already used across Re-ERP.

17. Validation Rules Catalog (per feature)

Every write service validates its payload and business state before touching data, and returns HTTP 400 (bad payload), 404 (referenced record missing), 409 (business rule violated) or 403 (no org access) with a specific message the UI shows verbatim. The catalog below is the implementation checklist per feature; screens duplicate the cheap checks client-side for instant feedback, but the server is always authoritative.

17.1 Organizations & parameters

Rule	Response
ORGANIZATION_CODE required, uppercase, unique	409 duplicate code
CHILD requires existing MASTER_ORG_ID and BUSINESS_UNIT_NAME from gl/businessunits	400 / 404
A MASTER org cannot reference a master; only one level of hierarchy	400
Deactivate blocked while org has on-hand, open shipments, open counts or open transfer orders	409 listing the blockers
Parameter upsert: accounts must be valid code combinations; PRIMARY_UOM must exist	400
Inter-org accounts required before the org can appear in a transfer	409 at transfer time naming the missing account
Receiving/shipping counters only increase; manual numbering rejects duplicates	409

17.2 Subinventories & locators

Rule	Response
SUBINVENTORY_CODE unique within org; required	409
Deactivate blocked while subinventory holds on-hand or open count entries	409
Locator create requires the subinventory to be PRESPECIFIED locator control	400
LOCATOR_CODE unique within subinventory	409
QUANTITY_TRACKED = N forces ASSET_FLAG = N (expense area)	400

17.3 Items, assignments, categories

Rule	Response
ITEM_NUMBER required, unique, no leading/trailing spaces	409
PRIMARY_UOM / item status / category values must exist and be ACTIVE	404
Serial + lot both enabled requires org enablement flags for both	400
Shelf life requires LOT_EXPIRATION_CONTROL = Y (and that requires LOT_CONTROL = Y)	400
Flag changes that conflict with existing stock are blocked (e.g. disabling lot control while lot on-hand exists; making item non-stockable while on-hand exists)	409
Assignment requires CHILD org and ACTIVE item; duplicate assignment rejected	409
Assignment deactivate blocked while item has on-hand or reservations in that org	409
One category per item per category set; MANDATORY sets enforced at item activation	409
Account/tax overrides must be valid combinations / existing tax codes	400

17.4 Lots & serials

Rule	Response
Lot details mandatory when item is lot controlled; sum of lot quantities = leg quantity	400
Expiration date mandatory on receipt when expiration controlled; defaulted from shelf life	400 if absent and no shelf life
Issues from an expired lot rejected unless reason EXPIRY is used	409
Lot material status must allow the action (4.9)	409
Serial numbers unique per item; a serial can only be issued from where it currently is; state machine IN_STOCK -> ISSUED / IN_TRANSIT enforced	409

17.5 Costing & cost periods

Rule	Response
An inventory org must be assigned to exactly one cost org before its first transaction	409 at transaction time
Item cost must be > 0; effective date not in a CLOSED period	400 / 409
Open period: only the period following the latest OPEN/CLOSED period	409

Rule	Response
Every transaction date must fall in an OPEN cost period and an open GL period	409
Close: zero unaccounted transactions (blocking), no open physical inventory (blocking), aged in-transit (warning only)	409 with blocker counts
Reopen only while the next period is not CLOSED	409

17.6 Transactions (engine, section 8.3)

Rule	Response
Org access (2.1) for the user	403
Item ACTIVE, transactable, with ACTIVE assignment in the org	409
Subinventory/locator exist, ACTIVE; material statuses of source and destination allow the action	409
Quantity > 0; UOM convertible to primary (item conversion else standard)	400
Negative legs: ATT check honoring reservations unless negative allowed (org + item)	409 listing available qty
Cost resolvable (item cost else standard cost)	409
Subinventory transfer: from and to differ, same org	400
Account alias must belong to the org and be ACTIVE	404
On-hand update and transaction insert are one database transaction - no partial writes	500 rolls back

17.6A Receiving stock - complete validation list (runs inside RR_INV_TXN_PKG.CREATE_TXN)

#	Validation	Failure
R0	Common gates 17.6 all pass first (access, org, type, item flags, assignment, subinventory/locator, material status, open cost + GL period, cost-org assignment, UOM conversion, reason, alias, no frozen PI, idempotency token)	403/400/404/409
R1	Unit cost, if entered, numeric and >= 0; REQUIRED on the first-ever receipt when the item has no cost basis and no standard cost	409
R2	Entered value (qty x cost) above configurable threshold -> warning / approval gate	warning
R3	Lot-controlled: lot details mandatory; sum of lot quantities equals transaction quantity	400
R4	New lot numbers only when org AUTO_CREATE_LOTS = Y, else the lot must pre-exist; existing lot must belong to the same item + org; number respects the item lot prefix	409
R5	Expiration-controlled: expiration date mandatory (defaulted from shelf life when configured) and later than the receipt date	400
R6	Serial-controlled: serial count equals quantity (integers only); serial unique per item; received serial must not be IN_STOCK anywhere (new, or ISSUED for return-type receipts)	409
R7	Destination material status allows receipt (QUARANTINE yes, REJECTED no)	409
R8	Resulting on-hand above org-assignment MAX_QTY -> warning (non-blocking)	warning
R9	Phase 2 PO receipts add: over-receipt tolerance %, early/late receipt days, substitute/unordered flags, receipt routing	reserved

17.6B Issuing stock - complete validation list (runs inside RR_INV_TXN_PKG.CREATE_TXN)

#	Validation	Failure
I0	Common gates 17.6 all pass first (same list as R0)	403/400/404/409
I1	Availability at the issue level (org + subinv + locator + lot): on-hand minus active reservations >= qty; a quoted demand reference adds back its own reservation - reserved stock is only issuable by its own demand. Error lists available qty and the blocking reservations	409

#	Validation	Failure
I2	Negative on-hand only when org NEGATIVE_INV_ALLOWED = Y AND the item allows it - NEVER for lot- or serial-controlled items	409
I3	Quoted demand reference: reservation exists, ACTIVE, same item/org, level-compatible; fulfilled qty updated atomically, auto-FULFILLED on completion	404/409
I4	Lot-controlled: lot details mandatory; sum equals qty; each lot has sufficient on-hand at that exact location	400/409
I5	Expired lot rejected unless reason = EXPIRY (deliberate scrap); lot material status allows issue; FEFO/FIFO suggested by the UI, not forced	409
I6	Serial-controlled: serial count equals qty; each serial IN_STOCK at that exact location; state moves to ISSUED in the same commit	409
I7	Cost must resolve (current average else standard cost) - no issue without a cost basis	409
I8	Offset account must resolve (alias -> item expense account -> org adjustment account); error names the missing account	409
I9	Resulting on-hand below MIN_QTY or SAFETY_STOCK -> warning (non-blocking), flagged to the min-max report	warning

Post-conditions asserted before COMMIT for both directions, rolled back on any failure: accounting lines balance (Dr = Cr); on-hand after = before +/- qty at every level touched; lot/serial masters consistent with the movement; audit stamps and the execution log written. Subinventory transfers, inter-org legs and count adjustments inherit these lists automatically because they execute through the same engine.

17.7 Reservations

Rule	Response
Item reservable; requested qty <= ATT at the requested level	409
Level consistency: locator/lot detail only where the matching control applies (locator control, lot control)	400
Cancel only ACTIVE reservations; fulfilled quantity is never reversed by cancel	409
Any stock-reducing action re-checks that remaining on-hand covers remaining active reservations	409 listing blocking reservations

17.8 Transfers & shipments

Rule	Response
From org differs from to org; both ACTIVE; item assigned and ACTIVE in both	400 / 409
Mode INTRANSIT requires in-transit accounts on both orgs; DIRECT requires inter-org receivable/payable	409 naming the missing account
Transfer order transitions only along DRAFT -> REQUESTED -> APPROVED -> IN_PROGRESS -> COMPLETED; CANCELLED only before first shipment	409
Ship quantity <= remaining requested; receive quantity <= shipped-not-received	409
Receive only shipments in SHIPPED status, only into the destination org, by a user with access to it	403 / 409

17.9 Counting & ABC

Rule	Response
Count entry only for stock in the count's org/scope; one open entry per item-location	409
Counted quantity >= 0; recount resets to PENDING keeping history	400
Approval required when variance exceeds tolerance; approver must differ from counter when approval required	403
Adjustment posts through the engine (all 17.6 rules apply, including period check)	as engine

Rule	Response
Physical inventory: snapshot freezes the scope - transactions in scope rejected while FREEZE_FLAG = Y	409
Tags: unique numbers, VOID needs a reason; completion requires every tag ENTERED or VOID	409
ABC compile requires on-hand or history for the criterion; classes must cover 100% of value	400

Cross-cutting: every table carries audit columns and services stamp them from the authenticated user; all lists are status-filtered by default with an explicit toggle to show inactive records; every mutation is logged with its payload in the standard API-transparency dialogs.

18. Design Validation - Executable Simulation

Before implementation, the documented rules were coded as a small in-memory engine (docs/design/inv_rd_simulation.py, committed with this document) and a realistic scenario was pushed through the full lifecycle. Every step asserts the invariants the design promises; the run passes end to end, demonstrating the design is internally consistent.

17.1 Invariants asserted

#	Invariant
I1	On-hand equals the signed sum of all transaction legs at every level (org/subinventory/item/lot)
I2	Every transaction's accounting lines balance (Dr = Cr) with the accounts from the type rules
I3	Available-to-transact = on-hand minus active reservations; reserved stock cannot be taken by unrelated issues
I4	Lot details always cover the leg quantity; per-lot on-hand reconciles
I5	Transactions are rejected outside an OPEN cost period
I6	Period close is blocked while unaccounted transactions exist; the snapshot equals on-hand x current cost
I7	In-transit quantity = shipped minus received, visible until receipt

17.2 Scenario and results (all steps passed)

Step	Action	Result
1	Opening receipt 100 EA FG-1001 into W1/STORES in lots L-2026-001 (60) and L-2026-002 (40)	On-hand 100; Dr Material 1,400.00 / Cr Adjustment 1,400.00
2	Reserve 25 EA for demand reference 'Project ALPHA'	ATT drops 100 -> 75
3	Unrelated issue of 90 EA	REJECTED: ATT 75 insufficient (reservation protected)
4	Issue 50 EA, reason DAMAGE, lot L-2026-001	On-hand 50; ATT 25
5	Issue 20 EA against 'Project ALPHA'	Reservation 20/25 fulfilled; on-hand 30
6	Subinventory transfer 10 EA STORES -> FG	STORES 20 / FG 10; net-zero accounting
7	In-transit ship 10 EA W1 -> W2, then receive at W2	In-transit 10 while SHIPPED, then W2 on-hand 10; Dr/Cr through in-transit account both sides
8	Cycle count: counted 17 vs snapshot 20	CYCLE_COUNT_ADJ -3; Dr Count-adjustment / Cr Material 42.00
9	Receipt dated 15-Aug-26 (period CLOSED)	REJECTED: cost period not OPEN
10	Close Sep-26 before accounting	REJECTED: 9 unaccounted transactions
11	Create Accounting (9 legs, all Dr=Cr), close Sep-26	CLOSED; snapshot W1/STORES 17 x 14.00 = 238.00, W2/STORES 10 x 14.00 = 140.00; total valuation 378.00 AED
12	Global reconciliation over all 9 legs	On-hand ledger identical to transaction sum at every level

The same scenario becomes the acceptance test for Phase 1c: after the real services are built, replaying these twelve calls against the ORDS endpoints must produce identical numbers.

19. Open Decisions

- Master organization: setup-only (recommended) or also transactable?
- Automatic Create Accounting after each transaction, or manual/periodic like the rest of Re-ERP? (Recommendation: manual button now, optional scheduled run later)
- Inter-org transfers across different cost orgs: keep at cost (Phase 1) - when is intercompany invoicing needed?
- Enable locators and lot control in the first rollout, or schema-ready but UI-hidden until a warehouse needs them?
- Cost periods: hard-block transactions when the GL period is open but the cost period is not (recommended), or warn only?
- Serial control: needed for any current business line, or defer the serial UI to Phase 2?